

Foundation (Jalapeno)

- Q1.** What is the purpose of a scatter plot in business?
- (A) To show stock prices over time
 - (B) To compare savings accounts
 - (C) To visualize the relationship between two variables
 - (D) To track tax rates
 - (E) To predict market trends
- Q2.** If the line of best fit has a positive slope, what does this indicate?
- (A) A negative correlation between variables
 - (B) No correlation between variables
 - (C) A positive correlation between variables
 - (D) A decrease in stock prices
 - (E) An increase in taxes
- Q3.** A company's profit P is related to the number of products sold x by the equation $P = 2x + 1000$. What is the y -intercept?
- (A) 1000
 - (B) 2000
 - (C) 0
 - (D) -1000
 - (E) -2000
- Q4.** What does a line of best fit enable us to do?
- (A) Make predictions about future data
 - (B) Identify patterns in past data
 - (C) Calculate the mean of a dataset
 - (D) Determine the range of a dataset
 - (E) Find the median of a dataset
- Q5.** A scatter plot shows a strong positive correlation between the amount of money invested x and the return on investment y . If x increases by 1000, what happens to y ?
- (A) Decreases by 100
 - (B) Increases by 100
 - (C) Remains the same
 - (D) Decreases by 1000
 - (E) Increases by 1000
- Q6.** The equation of the line of best fit for a dataset is $y = 2x - 3$. What is the slope of this line?
- (A) 1
 - (B) 2
 - (C) 3
 - (D) -1
 - (E) -2
- Q7.** A company's stock price y is related to the number of days x by the equation $y = -2x + 50$. What is the y -intercept?
- (A) 50
 - (B) 40
 - (C) 30
 - (D) 20
 - (E) 10
- Q8.** A scatter plot shows a weak negative correlation between the amount of money spent on advertising x and the company's profit y . If x increases by 500, what happens to y ?
- (A) Increases by 100
 - (B) Decreases by 100
 - (C) Remains the same
 - (D) Increases by 500
 - (E) Decreases by 500

Open Ended Questions (FRQ)

- Q9.** A company's daily profit P is related to the number of products sold x by the equation $P = 3x + 500$. If the company sells 20 products per day, what is the daily profit? Show your work and explain your answer.
- Q10.** A scatter plot shows a strong positive correlation between the amount of money invested x and the return on investment y . If the equation of the line of best fit is $y = 2x - 100$, and x increases by 500, what is the new return on investment? Show your work and explain your answer.

Chef's Tips